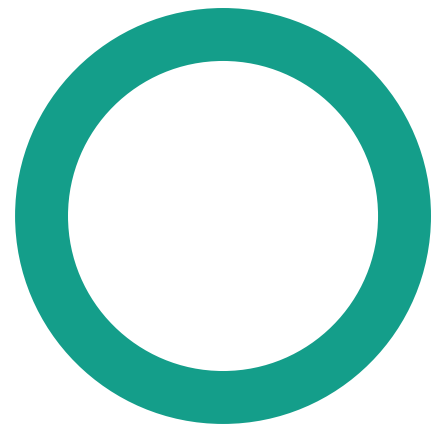


A STUDY SERIES HANDBOOK



MACHINE LEARNING

A Complete Beginner-Friendly Handbook

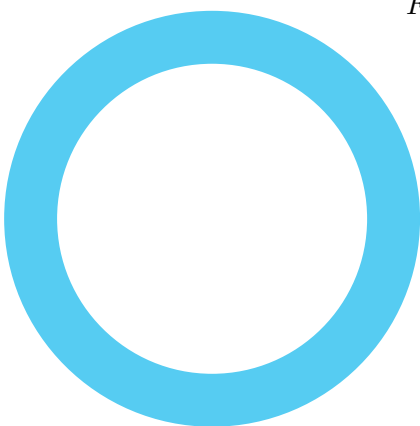
Foundations ■ AI vs. ML vs. DL ■ Types of Machine Learning

Foundations, Paradigms, and Applications

From learning from data to reinforcement learning

Adar's Study Series

Author: Adar Dey



How to Read This Book

This handbook develops the foundations of machine learning from first principles. Read one chapter at a time: each chapter introduces the core ideas in sequence and closes with a *Key Takeaways* box designed for rapid revision before examinations.

Three kinds of boxes recur throughout. Navy-headed boxes contain **core definitions** that are worth memorizing. Teal-headed boxes contain **worked examples and analogies** that ground the theory in practice. Amber boxes highlight **one-line intuitions** worth remembering verbatim.

In simple terms: machine learning is the process of learning from data.

Contents

How to Read This Book	2
1 Introduction to Machine Learning	1
1.1 What Is Machine Learning?	1
1.2 Machine Learning vs. Explicit Programming	1
1.2.1 Explicit (Traditional) Programming	1
1.2.2 The Machine Learning Paradigm	2
1.3 Why Do We Need Machine Learning?	2
1.3.1 Case 1 — Rules Are Impossible to Enumerate	2
1.3.2 Case 2 — The Space of Possible Cases Is Enormous	3
1.3.3 Case 3 — Discovering Hidden Structure (Data Mining)	3
1.4 A Brief History and the Drivers of Growth	3
1.5 Machine Learning and the Job Market	3
2 AI, Machine Learning, and Deep Learning	5
2.1 The Relationship Between AI, ML, and DL	5
2.2 Artificial Intelligence	6
2.2.1 What Constitutes Intelligence?	6
2.2.2 The Current State of AI	6
2.3 Early AI: Symbolic AI and Expert Systems	6
2.3.1 Limitations of Symbolic AI	7
2.4 Machine Learning in the AI Landscape	7
2.5 Deep Learning	7
2.5.1 Biological Inspiration	7
2.5.2 Automatic Feature Extraction	8
2.5.3 Hidden Layers and Representation Learning	8
2.5.4 The Effect of Data Size on Performance	8
2.6 Applications of Deep Learning	8
2.7 Should We Always Use Deep Learning?	9
2.8 Final Summary — The Umbrella Analogy	9
3 Types of Machine Learning: Supervised Learning	11
3.1 A Taxonomy by Amount of Supervision	11
3.1.1 Understanding “External Supervision”	11
3.2 Supervised Learning	12
3.2.1 Basic Concept	12
3.2.2 Why Is It Called “Supervised”?	12
3.2.3 Applications	12
3.3 Two Main Types of Supervised Learning	13
3.3.1 Regression	13
3.3.2 Classification	14

3.4	Regression vs. Classification	14
4	Unsupervised Machine Learning	16
4.1	Learning Without Labels	16
4.1.1	Basic Concept	16
4.1.2	The Goal of Unsupervised Learning	16
4.2	The Four Pillars of Unsupervised Learning	17
4.3	Clustering	17
4.4	Dimensionality Reduction	18
4.4.1	Why Is It Needed?	18
4.4.2	Feature Extraction	18
4.4.3	Data Visualization	18
4.5	Anomaly Detection	18
4.6	Association Rule Learning	19
4.7	Summary of Unsupervised Learning	20
5	Semi-Supervised and Reinforcement Learning	21
5.1	Semi-Supervised Learning	21
5.1.1	Why Do We Need It?	21
5.1.2	How Does It Work?	21
5.1.3	Advantages and Applications	22
5.2	Reinforcement Learning	22
5.2.1	Core Components	22
5.2.2	How Reinforcement Learning Works	22
5.2.3	Trial-and-Error Learning	24
5.2.4	Applications	24
5.3	Comparing the Four Paradigms and Final analogy - Four learning styles	25

Chapter 1

Introduction to Machine Learning

Machine learning is one of the most influential technologies of the modern era, powering systems that filter email, translate languages, diagnose disease, and drive cars. This chapter defines what machine learning is, contrasts it with traditional programming, and explains why it has become so important so quickly.

1.1 What Is Machine Learning?

◇ Definition — Machine Learning

A field of computer science that enables computer systems to **learn from data** using statistical techniques, **without being explicitly programmed for every task**. Instead of hand-writing detailed rules for every possible situation, the system automatically *discovers patterns* in data and uses them to make predictions or decisions.

1.2 Machine Learning vs. Explicit Programming

One of the most important conceptual steps in studying machine learning is understanding how it differs from traditional software development.

1.2.1 Explicit (Traditional) Programming

In traditional programming, the programmer writes the complete logic by hand. The workflow is

$$\text{Input} + \underbrace{\text{Program}}_{\text{hand-written rules / logic}} \longrightarrow \text{Output}.$$

Its defining characteristics are:

- every possible case must be handled by explicitly written code;
- the decision logic lives entirely with the developer;
- whenever a new situation appears, the program must be modified manually.

Example — Spam Detection (Rule-Based)

A traditional spam filter might contain rules such as

```
IF email contains "Huge Discount" THEN Spam.  
ELSE IF email contains "Free Money" THEN Spam.
```

This approach works only for rules the programmer anticipated. Any spam phrasing that was not foreseen slips through untouched.

1.2.2 The Machine Learning Paradigm

Machine learning inverts the workflow. Instead of writing the rules, we supply three ingredients:

- **input data**,
- **correct outputs** (labels), and
- **a machine learning algorithm**.

The algorithm automatically discovers the mathematical relationship (the patterns) between inputs and outputs:

$$\underbrace{\text{Input data} + \text{Correct outputs}}_{\text{examples}} + \text{Learning algorithm} \longrightarrow \text{Learned model (the logic)}.$$

The learned *model* is precisely the decision logic of the system — except that it was *discovered* from data rather than written by hand. Once training is complete, deployment is simple:

$$\text{New input} \longrightarrow \text{learned model} \longrightarrow \text{prediction}.$$

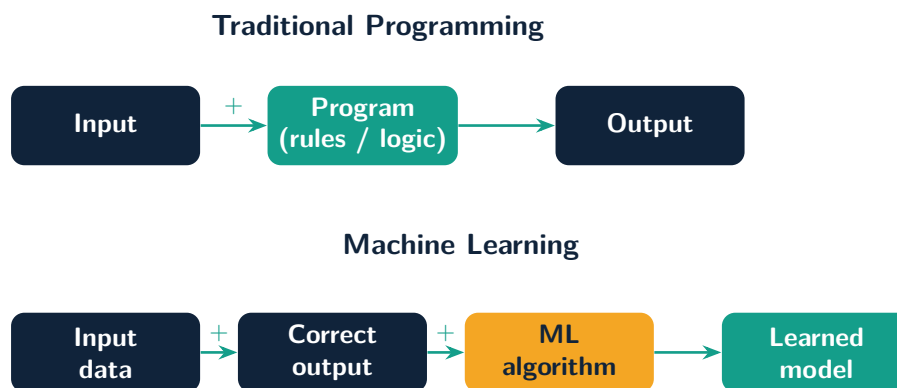


Figure 1.1: Explicit programming versus machine learning: in the former the human writes the rules; in the latter the rules are learned from data.

1.3 Why Do We Need Machine Learning?

Machine learning becomes essential precisely where traditional programming becomes difficult or impossible. Three scenarios illustrate this.

1.3.1 Case 1 — Rules Are Impossible to Enumerate

Example — Email Spam Classification

A developer builds a spam detector around keywords such as **Huge**, **Discount**, **Free**, and **Offer**. Spammers bypass these rules trivially with alternatives like **Big**, **Massive**, and **Special Offer**. Because language keeps changing, maintaining thousands of manual rules becomes impossible.

Machine learning solves this by learning spam patterns directly from data: whenever new training data is provided, the learned logic adapts automatically instead of requiring manual reprogramming.

1.3.2 Case 2 — The Space of Possible Cases Is Enormous

Example — Image Classification (Dogs)

Dogs come in hundreds of breeds with different sizes, colors, ear shapes, and body structures. Writing explicit rules covering every possible appearance is practically impossible. Machine learning instead learns the distinguishing characteristics automatically from thousands of labeled images — much as humans learn by observing many examples.

1.3.3 Case 3 — Discovering Hidden Structure (Data Mining)

Organizations accumulate enormous volumes of data. Traditional analysis with graphs and charts reveals only the trends that are visible on the surface, while many valuable relationships remain hidden deep inside large datasets. Machine learning algorithms extract these hidden patterns automatically — a process known as **data mining**.

1.4 A Brief History and the Drivers of Growth

Machine learning is not a new technology: it has existed for roughly **40–50 years**, yet it entered mainstream consciousness only after about **2010**. As with many quiet breakthroughs, decades of groundwork preceded sudden recognition. Two developments explain the explosion:

1. **Availability of massive data.** Machine learning requires large datasets, and the rise of the Internet, smartphones, social media, and online services generates enormous amounts of data every day. Abundant data has dramatically improved the performance of learning systems.
2. **Powerful hardware.** Earlier computers were severely limited — 128 MB of RAM was once considered impressive. Today, even smartphones carry 8–12 GB of RAM alongside GPUs capable of performing billions of mathematical operations in parallel. Modern hardware makes it practical to train large models; without it, many algorithms would simply be too slow to be useful.

1.5 Machine Learning and the Job Market

Machine learning professionals currently command high salaries. The primary reason is a *shortage of skilled talent*: when demand for ML engineers far exceeds the number of qualified professionals, companies compete for them, driving compensation upward.

Analogy — The Early Days of Java

When Java first became popular, experienced Java developers were scarce, so companies offered high salaries to attract them. Over time, as more people learned Java, the supply of developers increased and salaries normalized. The same dynamic is expected for machine learning: as more professionals enter the field, compensation will gradually stabilize.

Machine learning is still in its early growth phase — which makes **now** an excellent time to learn the subject.

★ Key Takeaways

- ✓ Machine learning lets systems learn patterns from data instead of following manually written rules.
- ✓ Traditional programming: $Input + Program \rightarrow Output$. Machine learning: $Input + Output + Algorithm \rightarrow Model$.
- ✓ ML is essential when rules are too complex, too numerous, or hidden deep inside large datasets.
- ✓ ML has existed for 40–50 years but exploded after 2010 thanks to massive data and powerful GPUs.
- ✓ High demand and a shortage of skilled professionals currently make ML a high-paying, high-opportunity field.

Chapter 2

AI, Machine Learning, and Deep Learning

Artificial intelligence, machine learning, and deep learning are closely related but *not* the same thing. This chapter situates machine learning inside the broader AI landscape and explains why deep learning transformed the field.

2.1 The Relationship Between AI, ML, and DL

The three fields nest inside one another:

Artificial Intelligence (AI) $\supset$ Machine Learning (ML) $\supset$ Deep Learning (DL).

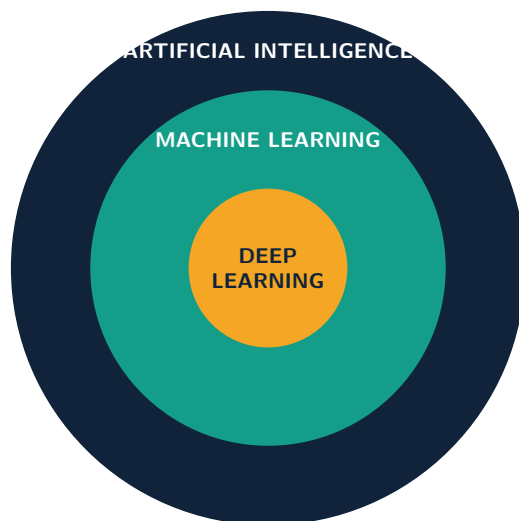


Figure 2.1: The nesting relationship: AI is the broadest field, ML is a subset of AI, and DL is a specialized subset of ML.

Every deep learning system is a machine learning system, and every machine learning system is an AI system. The converse does not hold: not every AI system uses machine learning, and not every machine learning system uses deep learning.

2.2 Artificial Intelligence

◇ Definition — Artificial Intelligence

A branch of computer science whose goal is to build machines that exhibit **human-like intelligence** — enabling machines to perform tasks that normally require human intelligence.

2.2.1 What Constitutes Intelligence?

Human intelligence is extraordinarily complex. It encompasses many distinct abilities:

- logical reasoning,
- problem-solving,
- learning,
- creativity,
- imagination,
- decision making,
- emotional intelligence,
- adaptation.

Building a machine that possesses *all* of these abilities remains one of the deepest open challenges in computer science.

2.2.2 The Current State of AI

Today's AI systems focus mainly on tasks that are measurable and well-defined. Researchers continue to work toward **Artificial General Intelligence (AGI)** — a system capable of performing almost any intellectual task a human can perform. Despite significant progress, a truly general-purpose intelligent system has not yet been achieved.

2.3 Early AI: Symbolic AI and Expert Systems

Before machine learning became popular, AI relied primarily on **Symbolic AI**, also known as **expert systems**. Rather than learning from data, developers manually encoded expert knowledge into the system as a knowledge base of numerous if-else rules.

Expert systems were built through a simple pipeline:

1. consult domain experts;
2. collect their knowledge;
3. convert that knowledge into logical rules;
4. apply those rules to solve problems.

Example — A Medical Expert System

A medical expert might state: “If the patient has symptom A, B, and C, then disease X is likely.” The programmer implements this knowledge directly as if-else statements.

Chess programs are the classic example of expert systems: because chess has fixed rules and a clearly defined environment, rule-based systems performed exceptionally well there.

2.3.1 Limitations of Symbolic AI

Expert systems work effectively only in *closed, well-defined domains* where the rules can be written down explicitly. They struggle badly with complex real-world problems.

Why Rules Fail — Image Classification

Suppose we want to determine whether an image contains a dog. Rules such as “dogs have four legs, two ears, and a tail” are hopelessly insufficient: there are hundreds of breeds with varying colors and sizes, and photographs differ in lighting and viewing angle. It is practically impossible to write rules covering every case.

This limitation directly motivated the rise of machine learning.

2.4 Machine Learning in the AI Landscape

Recall the definition from Chapter 1: machine learning is the branch of computer science that uses statistical techniques to discover patterns within data. Within the AI landscape, it is precisely the subset of AI in which — instead of programming every rule manually — the algorithm automatically *learns* the relationship between inputs and outputs.

Concretely, we provide input data, correct outputs (labels), and a learning algorithm. The algorithm studies the data and learns the mathematical relationship between inputs and outputs; this learned relationship becomes the **model**. Once training is complete, the model predicts outputs for new, unseen inputs. Machine learning is fundamentally an engineering discipline built upon **statistics and mathematics**, with the objective of extracting meaningful patterns from data and using them for prediction and decision making.

Example — Dog Recognition

Suppose we provide thousands of images, each labeled **Dog** or **Not Dog**. The algorithm analyzes these examples and gradually discovers the hidden characteristics that distinguish dogs from other objects. These learned characteristics form the *underlying representation* of the data — and this automatic discovery process is exactly what we call **learning**.

2.5 Deep Learning

◇ Definition — Deep Learning

A specialized branch of machine learning. Like traditional ML, it learns from data, trains models using algorithms, and makes predictions on unseen data — but the algorithms it employs are fundamentally different: deep artificial neural networks.

2.5.1 Biological Inspiration

Deep learning algorithms are loosely inspired by the structure of the human brain. Their basic computational unit is the **artificial neuron**, historically called the **perceptron**. Although artificial neurons do not function exactly like biological neurons, their design is modeled on the brain’s architecture. A network containing many such neurons is an **Artificial Neural Network (ANN)**.

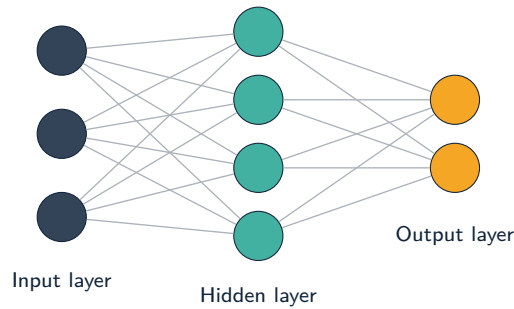


Figure 2.2: A small artificial neural network. Each circle is an artificial neuron (perceptron); successive layers transform raw inputs into increasingly abstract internal representations.

2.5.2 Automatic Feature Extraction

The main limitation of traditional machine learning is that it usually requires **feature engineering**.

◇ Definition — Feature

A **feature** is a measurable property or characteristic of the data. For example, when distinguishing dogs from cats, useful features might include ear shape, face shape, tail length, and body structure.

In traditional ML, features are designed manually by human experts. This *feature engineering* is difficult, time-consuming, and highly domain-specific; for many complex problems, even deciding *which* features should be used is extremely challenging.

Deep learning removes this bottleneck. Instead of relying on hand-designed features, it learns useful features directly from raw data — a process known as **automatic feature extraction**. As a result, deep learning performs exceptionally well on highly complex tasks where manually identifying useful features is difficult or impossible.

2.5.3 Hidden Layers and Representation Learning

Deep learning models contain multiple layers of artificial neurons, called **hidden layers**. Each layer learns increasingly complex representations of the data. In general, more layers allow the model to learn more abstract and sophisticated patterns, often improving accuracy on complex problems. This ability to discover hidden representations is one of deep learning's greatest strengths.

2.5.4 The Effect of Data Size on Performance

Traditional machine learning models tend to *plateau*: beyond a certain point, adding more training data yields only limited improvement. Deep learning behaves differently — as the amount of training data increases, deep models typically keep improving. This scaling behavior is one of the central reasons for deep learning's recent success.

2.6 Applications of Deep Learning

Deep learning has achieved remarkable success in areas such as:

- image classification,
- object detection,

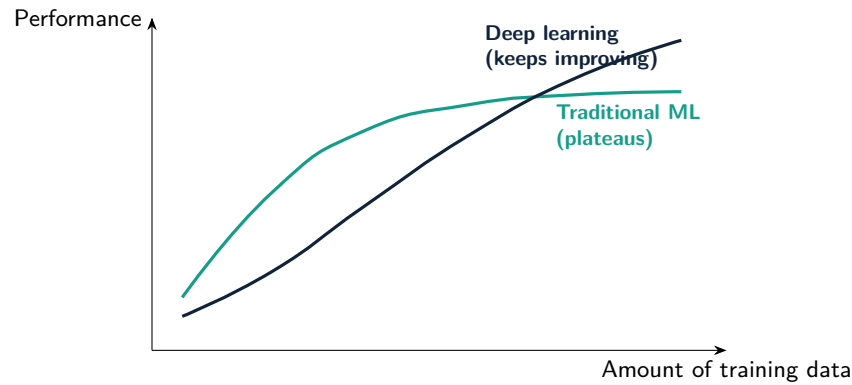


Figure 2.3: Performance versus training data: traditional ML methods tend to plateau, whereas deep learning models typically keep improving as more data becomes available.

- face recognition,
- speech recognition,
- natural language processing (NLP),
- text analysis,
- machine translation.

On many of these tasks, deep learning significantly outperforms traditional machine learning methods.

2.7 Should We Always Use Deep Learning?

No. Deep learning is not automatically the best choice for every problem.

Many organizations — especially in banking, insurance, and similar industries — do not possess massive amounts of *labeled* data. In such situations, traditional machine learning algorithms often perform better and are far more practical. The choice between ML and DL should depend on:

- data availability,
- problem complexity,
- computational resources,
- business requirements.

2.8 Final Summary — The Umbrella Analogy

Analogy — An Umbrella

The umbrella itself represents **Artificial Intelligence** — the overarching goal of creating intelligent machines.

The main fabric represents **Machine Learning** — the primary approach enabling machines to learn from data using statistical methods.

The finely woven inner fibers represent **Deep Learning** — a specialized technique

that automatically extracts complex features and hidden patterns from data.

★ Key Takeaways

- ✓ AI is the broadest goal; ML is a subset of AI; DL is a specialized subset of ML.
- ✓ Early AI used symbolic AI / expert systems (hand-written if-else rules) — strong in closed domains like chess, weak on messy real-world data.
- ✓ ML learns the input→output relationship statistically from data instead of manual rules.
- ✓ DL is brain-inspired (artificial neurons / ANNs) and performs automatic feature extraction using hidden layers.
- ✓ DL keeps improving with more data, unlike traditional ML which plateaus — but DL is not the right choice when labeled data is scarce.

Chapter 3

Types of Machine Learning: Supervised Learning

Machine learning algorithms can be classified by *how much external supervision they receive during training*. The amount of supervision determines how the algorithm learns from the available data. This chapter develops the first and most widely used category: supervised learning.

3.1 A Taxonomy by Amount of Supervision

There are four major categories of machine learning:

- **Supervised learning,**
- **Unsupervised learning,**
- **Semi-supervised learning,**
- **Reinforcement learning.**

These four categories constitute the most widely accepted classification of machine learning algorithms.

3.1.1 Understanding “External Supervision”

External supervision refers to the guidance provided to a machine learning algorithm during training — usually in the form of correct output labels.

Example — A Student Placement Dataset

Suppose we want to build a model that predicts whether a student will be placed:

Table 3.1: A labeled dataset for placement prediction. IQ and CGPA are the input features; Placement is the target label.

IQ	CGPA	Placement
120	3.80	Yes
115	3.50	Yes
98	2.70	No
130	3.95	Yes

Here **IQ** and **CGPA** are the *input features*, and **Placement** is the *output label*. If the correct labels are available, the algorithm receives supervision; if they are absent, the algorithm must discover patterns on its own. The availability of labels is the dividing line between the four categories of machine learning.

3.2 Supervised Learning

◇ Definition — Supervised Learning

A type of machine learning in which the algorithm is trained on a dataset containing **both input data and the correct output (labels)**. Because the correct answers are already available, the algorithm learns the relationship between inputs and outputs, aiming to predict accurately on new, unseen data.

3.2.1 Basic Concept

A supervised dataset always contains **input features** X and a **target output** Y . Given labeled examples $\{(x^{(i)}, y^{(i)})\}_{i=1}^n$, the algorithm learns a mathematical function $f : \mathcal{X} \rightarrow \mathcal{Y}$ that maps inputs to correct outputs. After training, the learned model can predict the output for previously unseen inputs.

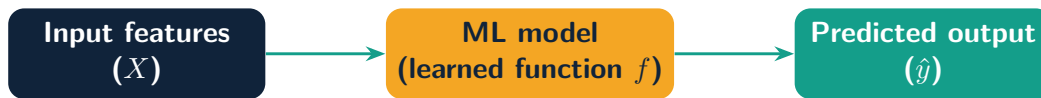


Figure 3.1: The supervised learning pipeline: a model learns a mapping from input features to target outputs from labeled examples.

Example — Placement Prediction

A company possesses historical placement data (IQ, CGPA, placement status). The ML algorithm learns the relationship between IQ, CGPA, and placement. Later, given the IQ and CGPA of a *new* student, the trained model predicts whether that student is likely to be placed.

3.2.2 Why Is It Called “Supervised”?

The name is literal: the algorithm learns *under the guidance of known correct answers* — much like a student learning under a teacher who already knows the correct solution and gives continuous feedback.

3.2.3 Applications

Supervised learning is the most commonly used type of machine learning. It is widely applied in:

- spam email detection,
- house price prediction,
- medical diagnosis,
- credit risk assessment,

- weather prediction,
- stock market forecasting,
- customer churn prediction,
- fraud detection,
- student placement prediction.

Most real-world machine learning applications belong to this category.

3.3 Two Main Types of Supervised Learning

Supervised problems are classified according to the *type of output variable*. There are two major types: **regression** and **classification**.

3.3.1 Regression

◇ Definition — Regression

A supervised learning problem in which the output variable is **numerical**: the objective is to predict a continuous value. In symbols, the target satisfies $y \in \mathbb{R}$.

Characteristics:

- the output is a number;
- the predicted value can take many possible values within a range;
- used whenever the target variable represents a quantity or measurement.

Examples — Regression in Practice

Predicting house prices, employee salaries, a student's expected salary package, tomorrow's temperature, sales revenue, or rainfall. Given IQ and CGPA, a model may output **Salary** = 4.5 or **Salary** = 6.2 (Lakh). Since the output is numerical, this is a regression problem.

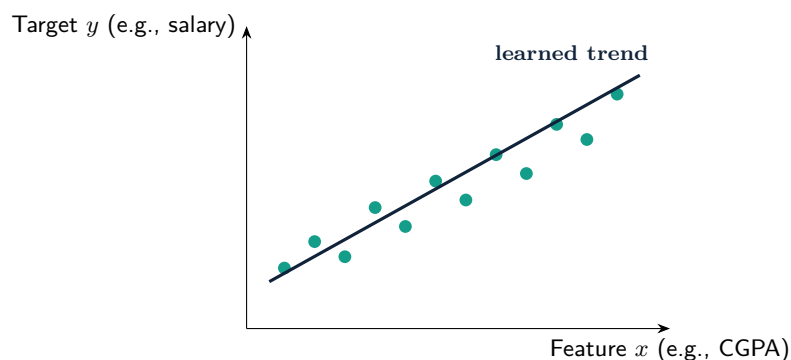


Figure 3.2: Regression predicts a continuous numerical target; the fitted line summarizes the learned input–output relationship.

3.3.2 Classification

◇ Definition — Classification

A supervised learning problem in which the output variable belongs to one of several **predefined categories**. Instead of predicting numbers, the algorithm predicts *class labels*. In symbols, the target satisfies $y \in \{c_1, c_2, \dots, c_k\}$.

Characteristics:

- the output is categorical;
- every prediction belongs to a predefined class;
- used whenever the target variable represents categories.

Examples — Classification in Practice

Spam or not spam; yes or no; pass or fail; disease or no disease; cat or dog; genuine or fraudulent transaction.

Given IQ and CGPA, the model predicts **Placement** = **Yes** or **No**. Because the output belongs to one of two categories, this is a classification problem. Likewise, “will it rain tomorrow?” is classification, since the output is **Rain** or **No Rain** — not a number.

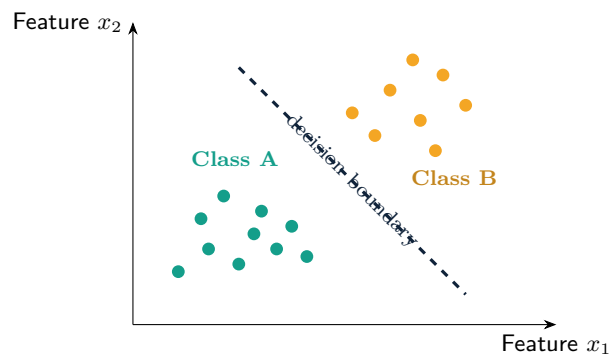


Figure 3.3: Classification predicts a discrete class label; a decision boundary separates the regions assigned to each class.

3.4 Regression vs. Classification

Table 3.2: Regression versus classification at a glance.

Regression	Classification
Predicts numerical values	Predicts categorical values
Output is continuous (any value in a range)	Output belongs to a fixed set of classes
Example: house price	Example: spam detection
Example: salary prediction	Example: disease detection
Example: temperature prediction	Example: placement prediction

★ Key Takeaways

- ✓ ML has four categories based on how much supervision (labeled data) is available: supervised, unsupervised, semi-supervised, reinforcement.
- ✓ Supervised learning trains on labeled data (input features X + target output Y) to learn a predictive mapping.
- ✓ Regression predicts numerical, continuous values (price, salary, temperature).
- ✓ Classification predicts categorical class labels (spam / not spam, pass / fail).
- ✓ Supervised learning is the most widely used ML approach in real-world applications.

Chapter 4

Unsupervised Machine Learning

Not every dataset comes with answers. This chapter studies the second pillar of machine learning: learning structure from *unlabeled* data.

4.1 Learning Without Labels

◇ Definition — Unsupervised Learning

A type of machine learning in which the algorithm is trained **without labeled data**. The dataset contains only input features and no output labels. Since there are no target labels, the algorithm cannot learn a direct input–output relationship; instead, its objective is to discover **hidden patterns, structures, and relationships** within the data.

4.1.1 Basic Concept

- only input data (features) are available;
- no target output (label) is provided;
- the algorithm explores the data independently;
- it identifies similarities, differences, structures, and hidden relationships among data points.

Example — A Student Dataset Without Labels

Consider a student dataset containing only IQ and CGPA — notice that there is **no Placement column**. Since the algorithm does not know whether these students were placed, it cannot predict placement. Instead, it analyzes the dataset and discovers natural groups or hidden patterns among the students. This is the essence of unsupervised learning.

4.1.2 The Goal of Unsupervised Learning

The primary objective is *not* prediction. Instead, the goal is to:

- discover hidden patterns;
- identify natural groupings;
- understand the structure of the data;
- detect unusual observations;
- find relationships among variables or items.

4.2 The Four Pillars of Unsupervised Learning



Figure 4.1: The four pillars of unsupervised learning.

4.3 Clustering

◇ Definition — Clustering

The process of dividing data into groups (*clusters*) based on similarity: data points within the same cluster are more similar to each other than to data points in other clusters.

Imagine plotting all data points on a graph: the algorithm observes that some points naturally lie close together and groups them into the same cluster automatically, without any human guidance.

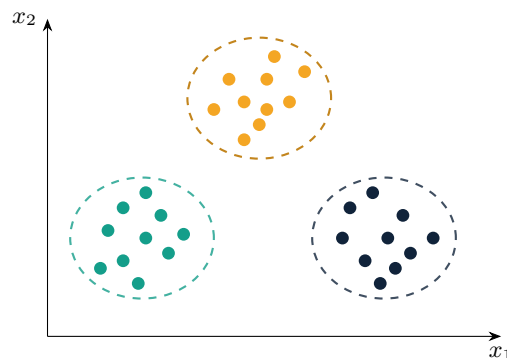


Figure 4.2: Clustering groups unlabeled points by similarity; the three natural groups were found without any labels.

Examples — Clustering in Action

A college may automatically group students by academic performance, attendance, study habits, and activity participation — without any predefined categories.

An e-commerce company can group customers by purchasing behavior, shopping frequency, spending habits, and product preferences — enabling personalized marketing strategies.

Clustering supports customer segmentation, student categorization, market analysis, recommendation systems, and pattern discovery — and is especially valuable for high-dimensional datasets where meaningful groups are not obvious to humans.

4.4 Dimensionality Reduction

◇ Definition — Dimensionality Reduction

The process of reducing the **number of input features (dimensions)** while preserving as much useful information as possible.

4.4.1 Why Is It Needed?

Modern datasets may contain hundreds or thousands of features. Too many input features create several problems:

- increased computational cost,
- slower training,
- higher memory usage,
- increased model complexity,
- greater risk of overfitting.

Moreover, adding features does not always improve performance — some features may be redundant or highly correlated.

4.4.2 Feature Extraction

One common approach is **feature extraction**: instead of simply deleting features, the algorithm combines multiple related features into a new feature that captures the essential information.

Example — Combining Features

A housing dataset contains “Number of Rooms” and “Number of Bathrooms” — both closely related to the overall size of the house. A dimensionality reduction technique may combine them into a new feature, such as **House Size (Square Footage)**, summarizing the important information while reducing complexity.

4.4.3 Data Visualization

High-dimensional data cannot be visualized directly, because humans interpret only two- and three-dimensional spaces. Dimensionality reduction transforms high-dimensional datasets into lower dimensions, making visualization — and the study of hidden structures — possible.

The benefits: faster model training, reduced computational cost, improved visualization, less noise, better generalization, and easier interpretation of the data.

4.5 Anomaly Detection

◇ Definition — Anomaly Detection

The process of identifying unusual or abnormal observations that differ significantly from the majority of the data. Such unusual observations are called **outliers** or **anomalies**.

The algorithm first learns what “normal” data looks like. When a new observation deviates significantly from this normal pattern, it is flagged as an anomaly — like a dense cluster of points on a graph with one far-away point that looks suspicious.

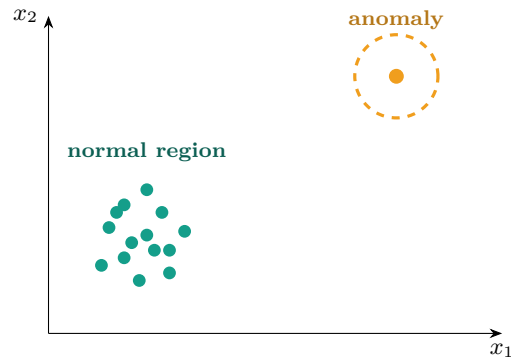


Figure 4.3: Anomaly detection learns the dense “normal” region and flags far-away observations as suspicious.

Applications include:

- credit card fraud detection,
- banking fraud prevention,
- loan approval systems,
- manufacturing defect detection,
- cybersecurity,
- network intrusion detection,
- medical diagnosis.

Example — Credit Card Fraud

A customer normally spends money in only one city. Suddenly, within a few minutes, transactions occur in another country. This behavior differs significantly from the customer’s normal spending pattern, so the anomaly detection system immediately flags the transaction as potentially fraudulent.

4.6 Association Rule Learning

◇ Definition — Association Rule Learning

A technique for discovering **hidden relationships or associations** among items in a dataset. Rather than predicting outcomes, it identifies which items frequently occur together.

The most common application is **Market Basket Analysis**: retail stores analyze customer purchase histories to determine which products are frequently bought together, helping optimize product placement and marketing strategies.

Example — Milk and Eggs

A supermarket analyzes 100 shopping bills. It observes that milk appears in many transactions, and whenever milk is purchased, eggs are also purchased in a large number of cases. This indicates a strong association between milk and eggs — so the supermarket may place these products close together to encourage more sales.

Association rule learning is also used for product recommendation systems, online shopping suggestions, inventory management, cross-selling strategies, and customer behavior analysis.

4.7 Summary of Unsupervised Learning

Table 4.1: The four unsupervised techniques and their primary purposes.

Technique	Primary Purpose
Clustering	Group similar data points together based on similarity
Dimensionality reduction	Reduce the number of features while preserving the important information
Anomaly detection	Detect abnormal or suspicious observations
Association rule learning	Discover relationships among items that frequently co-occur

★ Key Takeaways

- ✓ Unsupervised learning uses only input data — no output labels are provided.
- ✓ Since no correct answers exist, the algorithm learns independently to discover hidden structure.
- ✓ Four major applications: clustering, dimensionality reduction, anomaly detection, and association rule learning.
- ✓ Unlike supervised learning, it is used primarily for exploration and pattern discovery rather than direct prediction.

Chapter 5

Semi-Supervised and Reinforcement Learning

The final two paradigms complete the taxonomy: learning from a mixture of labeled and unlabeled data, and learning through interaction and reward.

5.1 Semi-Supervised Learning

◇ Definition — Semi-Supervised Learning

A machine learning paradigm that combines supervised and unsupervised learning. The training dataset contains a **small amount of labeled data** and a **large amount of unlabeled data**; the algorithm uses both to build a better predictive model.

5.1.1 Why Do We Need It?

In many real-world applications, collecting raw data is easy, but **labeling** it is expensive, time-consuming, and labor-intensive: medical images must be labeled by expert doctors, satellite imagery requires domain experts, speech datasets require manual transcription, and face datasets require identifying each individual. Semi-supervised learning reduces this burden by requiring only a small portion of the data to be labeled.

5.1.2 How Does It Work?

1. Collect a large dataset.
2. Label only a small number of samples.
3. Leave the remaining samples unlabeled.
4. Train the algorithm using both labeled and unlabeled data.
5. The algorithm learns from the labeled examples and exploits the hidden structure of the unlabeled data to improve.
6. In many cases, it can automatically assign labels to the remaining unlabeled data.

Thus, semi-supervised learning leverages the strengths of both supervised and unsupervised learning.

Walkthrough — Google Photos

Step 1 (unsupervised): Google Photos analyzes thousands of images and automatically groups photographs that appear to contain the same person — without knowing who that person is.

Step 2 (supervised): The app asks the user a simple question: “Who is this person?” The user provides the correct name only once.

Step 3 (automatic labeling): Having received this single label, the system automatically applies the same name to every other image in that cluster. This combination of clustering plus limited human labeling makes the overall system semi-supervised.

GOOGLE PHOTOS: SEMI-SUPERVISED FACE LABELING

Figure 5.1: Semi-supervised labeling in Google Photos: unsupervised clustering plus a single human label labels the entire cluster.

5.1.3 Advantages and Applications

- requires far fewer labeled examples;
- reduces annotation cost;
- saves time and human effort;
- improves performance compared to using only a small labeled dataset;
- makes use of abundant unlabeled data.

Semi-supervised learning is commonly applied in image recognition, face recognition, speech recognition, medical image analysis, text classification, document categorization, and web page classification.

5.2 Reinforcement Learning**◇ Definition — Reinforcement Learning**

A machine learning paradigm in which an **agent** learns by **interacting with its environment**. Unlike supervised learning, the agent is not given a labeled dataset at the start; it learns through **trial and error**, receiving *rewards* for good actions and *penalties* for poor actions. The goal is to learn the best strategy (*policy*) for maximizing long-term rewards.

5.2.1 Core Components**5.2.2 How Reinforcement Learning Works**

- the agent observes the environment;
- it chooses an action;
- the environment responds;

Table 5.1: The six core components of reinforcement learning.

Component	Description
Agent	The learner and decision-maker: observes the environment, takes actions, receives feedback, and improves over time (a robot, a game-playing AI, a self-driving car, an autonomous drone).
Environment	The world in which the agent operates (a chess board, a road, a video game, a robotic workspace); it responds to every action the agent takes.
Actions	Any decision made by the agent (move left or right, accelerate, brake, pick up or place an object); each action changes the state of the environment.
Reward	Positive feedback for a desirable action (winning a game, reaching the destination safely, completing a task efficiently); rewards encourage repeating successful behaviors.
Punishment	A penalty (negative reward) for an undesirable action (losing a game, crashing, hitting an obstacle, an illegal move); punishments discourage poor decisions.
Policy	The strategy that determines which action to take in a given situation; as the agent gains experience, it continuously updates and improves its policy.

- the agent receives a reward or punishment;
- the agent updates its policy;
- the cycle repeats many times.

Formally, the agent seeks a policy π that maximizes the expected cumulative reward,

$$\max_{\pi} \mathbb{E}[r_1 + r_2 + r_3 + \dots],$$

learning over time to maximize total reward while minimizing penalties.

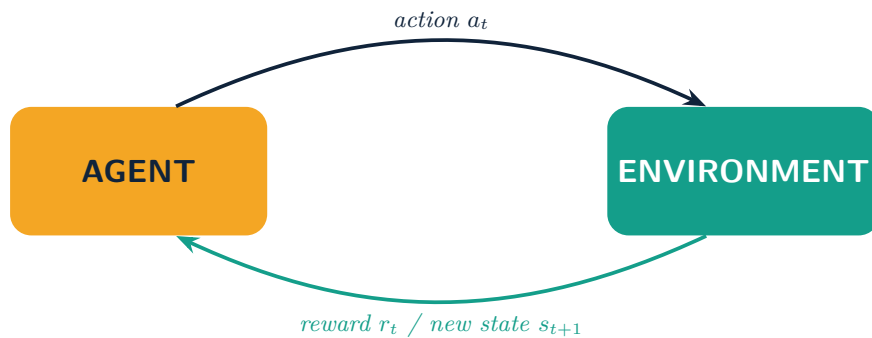


Figure 5.2: The agent–environment interaction loop of reinforcement learning; after every step the agent updates its policy.

5.2.3 Trial-and-Error Learning

The key idea behind reinforcement learning is learning through trial and error. Initially the agent has little or no knowledge — it explores different actions, makes mistakes, receives feedback, and gradually discovers which actions produce the best long-term results.

Example 1 — A Child

A young child learns many everyday activities by experimenting. Correct actions are encouraged through praise or rewards; incorrect actions are discouraged through correction. Over time, the child develops better behavior through experience.

Example 2 — Dog Training

When training a dog, good behavior is rewarded with treats and bad behavior is discouraged through correction. Gradually, the dog learns which actions are desirable — reinforcement learning follows exactly the same principle.

5.2.4 Applications

Reinforcement learning is applied in self-driving cars, robotics, video game AI, industrial automation, resource management, recommendation systems, and autonomous navigation.

Case Study — AlphaGo

One of the most famous applications of reinforcement learning is **AlphaGo**, developed by **DeepMind**. AlphaGo learned to play **Go**, one of the most complex games ever created — far more complex than chess in terms of possible positions. After extensive training, including playing against itself millions of times, AlphaGo defeated the world champion Lee Sedol in 2016. This was a landmark moment for AI: Go was long believed to be too intuitive and complex for computers to master, and AlphaGo's victory demonstrated how far trial-and-error learning can go.

5.3 Comparing the Four Paradigms and Final analogy - Four learning styles

Table 5.2: The four types of machine learning side by side.

Type	Training Data	Main Objective	Typical Applications	Applications
Supervised	Labeled data (input + output)	Learn the input→output mapping for prediction	Spam detection, price prediction, medical diagnosis	
Unsupervised	Unlabeled data	Discover hidden patterns and structures	Clustering, anomaly detection, market-basket analysis	
Semi-supervised	Small labeled + large unlabeled dataset	Improve learning while reducing labeling effort	Google Photos, medical imaging, face recognition	
Reinforcement	No predefined dataset; learned through interaction	Maximize cumulative rewards via the best policy	Robotics, autonomous vehicles, game-playing AI	

Table 5.3: The four paradigms as human learning styles.

Type	Analogy
Supervised	A student learns under the guidance of an experienced teacher who already knows the correct answers.
Unsupervised	An archaeologist discovers ancient artifacts without labels, analyzing similarities to find hidden patterns.
Semi-supervised	A teacher answers only a few examples; the student infers the rest using those plus many unanswered questions.
Reinforcement	A newborn child learns by interacting with the world — through rewards, corrections, and repeated experience.

The four categories of machine learning differ primarily in how much supervision is available during training — almost every machine learning problem can be approached using one of these four learning strategies.

★ Key Takeaways

- ✓ Semi-supervised learning combines a small labeled set with a large unlabeled set — ideal when labeling is expensive (e.g. Google Photos).
- ✓ Reinforcement learning has six core components: agent, environment, actions, reward, punishment, and policy.
- ✓ RL learns through trial and error, similar to how a child or a dog learns from consequences.
- ✓ AlphaGo (by DeepMind) is a landmark RL success story, mastering the complex game of Go.
- ✓ All four ML types — supervised, unsupervised, semi-supervised, reinforcement — differ mainly in how much labeled supervision they receive.